Quasi-symmetric refinement of the cylindric partition generating function

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Abstract

We show that the generating function for rank-3 cylindric plane partitions, refined by value frequencies, is a quasi-symmetric function. Specifically, for a profile $c = (c_0, c_1, c_2)$, the power series $\widetilde{\mathrm{GK}}_c(\mathbf{z}) = \sum_{\pi \in \mathrm{CPP}(c)} z_1^{\#_1(\pi)} z_2^{\#_2(\pi)} \cdots$ (where $\#_k(\pi)$ counts the number of parts equal to k in π) lies in the ring QSym of quasi-symmetric functions. This is verified computationally for all profiles with $d = |c| \leq 5$. Under the principal specialization $z_k \mapsto q^k$, this recovers $\mathrm{GK}_c(1, q) = \sum_n g_n(q)$. Moreover, the expansion in Gessel's fundamental basis $\{L_\alpha\}$ has nonneg coefficients (L-positivity), a strictly stronger property than M-positivity. However, the quasi-symmetry applies only to the *unconstrained* generating function (all CPPs, no bound on the maximum entry). Since Warnaar's $Q_{n,c}(q)$ polynomials are extracted from the max $= n$ piece, the QSym structure does not directly bear on the positivity conjecture.

1 Setup

A *rank-3 cylindric plane partition* with profile $c = (c_0, c_1, c_2)$ is a triple of partitions $\pi = (\nu^{(1)}, \nu^{(2)}, \nu^{(3)})$ satisfying the shifted dominance (interlacing) conditions:

$$\nu_j^{(1)} \geq \nu_{j+c_1}^{(2)}, \qquad \nu_j^{(2)} \geq \nu_{j+c_2}^{(3)}, \qquad \nu_j^{(3)} \geq \nu_{j+c_0}^{(1)}, \tag{1}$$

for all $j \geq 0$, where $\nu_j^{(i)} = 0$ for $j \geq \ell(\nu^{(i)})$.

The two-variable generating function is

$$\mathrm{GK}_c(z, q) = \sum_{\pi \in \mathrm{CPP}(c)} z^{n(\pi)}\, q^{|\pi|} = \sum_{n \geq 0} g_n(q)\, z^n,$$

where $n(\pi) = \max\{\nu_j^{(i)} : i, j\}$ is the maximum entry (equivalently, the number of paths in the NILP model) and $|\pi| = |\nu^{(1)}| + |\nu^{(2)}| + |\nu^{(3)}|$ is the total weight.

Warnaar's Conjecture 2.7 asserts that the polynomials

$$Q_{n,c}(q) = (q;q)_n \cdot [z^n]\big((zq;q)_\infty \cdot \mathrm{GK}_c(z,q)\big)$$

have nonneg integer coefficients for all $n \geq 0$ and all profiles c with $d = |c| \not\equiv 0 \pmod 3$.

2 The frequency refinement

For a cylindric partition $\pi = (\nu^{(1)}, \nu^{(2)}, \nu^{(3)})$, define the *value frequency* of k as

$$\#_k(\pi) = \big|\{(i,j) : \nu_j^{(i)} = k\}\big|,$$

the total number of parts equal to k across all three partitions. The weight decomposes as $|\pi| = \sum_{k\geq 1} k \cdot \#_k(\pi)$.

Definition 1 (Frequency generating function)**.** *For a profile c, define*

$$\widetilde{\mathrm{GK}}_c(\mathbf{z}) = \sum_{\pi \in \mathrm{CPP}(c)} z_1^{\#_1(\pi)}\, z_2^{\#_2(\pi)} \cdots z_n^{\#_n(\pi)} \cdots \;\in\; \mathbb{Z}[[z_1, z_2, \ldots]], \qquad (2)$$

where the sum is over all cylindric partitions with profile c (of any maximum entry).

Remark 2. *Under the specialization $z_k \mapsto q^k$, we recover*

$$\widetilde{\mathrm{GK}}_c(q, q^2, q^3, \ldots) = \sum_{\pi} q^{|\pi|} = \mathrm{GK}_c(1, q).$$

The maximum entry $n(\pi) = \max\{k : \#_k(\pi) > 0\}$ is recoverable from the frequency data but is not a linear statistic, so $z^{n(\pi)}$ is not a monomial specialization of the z_k.

3 Quasi-symmetry

Recall that a formal power series $f \in \mathbb{Z}[[z_1, z_2, \ldots]]$ is *quasi-symmetric* if for every composition $\alpha = (\alpha_1, \ldots, \alpha_k)$ with $\alpha_i > 0$, the coefficient of $z_{i_1}^{\alpha_1} \cdots z_{i_k}^{\alpha_k}$ is the same for all choices of indices $1 \leq i_1 < i_2 < \cdots < i_k$.

The ring QSym of quasi-symmetric functions sits between the ring $\mathrm{Sym} = \Lambda$ of symmetric functions and the full power series ring:

$$\mathrm{Sym} \ \subset \ \mathrm{QSym} \ \subset \ \mathbb{Z}[[z_1, z_2, \ldots]].$$

The standard monomial basis of QSym is $\{M_\alpha\}_\alpha$, indexed by compositions, where

$$M_\alpha = \sum_{i_1 < i_2 < \cdots < i_k} z_{i_1}^{\alpha_1} z_{i_2}^{\alpha_2} \cdots z_{i_k}^{\alpha_k}.$$

Observation 3 (Computational)**.** *For every rank-3 profile c with $d = |c| \leq 5$ (55 profiles total), $\widetilde{\mathrm{GK}}_c(\mathbf{z})$ is quasi-symmetric. That is,*

$$\widetilde{\mathrm{GK}}_c(\mathbf{z}) = \sum_\alpha a_\alpha(c)\, M_\alpha, \tag{3}$$

with all coefficients $a_\alpha(c) \in \mathbb{Z}_{\geq 0}$.

The nonnegativity of the coefficients $a_\alpha(c)$ is automatic (they count cylindric partitions), so the content of the observation is the quasi-symmetry itself: the count depends only on the *composition* of value frequencies, not on which values carry those frequencies.

Remark 4. $\widetilde{\mathrm{GK}}_c$ *is* not *symmetric. For example, with $c = (1, 1, 0)$, the coefficient of $M_{(2,1)}$ is 4 while the coefficient of $M_{(1,2)}$ is 3. Since $(2, 1)$ and $(1, 2)$ are different compositions (related by rearrangement, hence giving the same partition), symmetry would require these to be equal.*

4 Monomial QSym coefficients for small profiles

4.1 Profile $c = (1, 1, 0)$, $d = 2$

The QSym coefficients $a_\alpha = a_\alpha(1, 1, 0)$ display a remarkable stabilization: once all parts of the composition α are ≥ 2, the coefficient depends only on the length of α.

Composition α	a_α
()	1
(k), any $k \geq 1$	2
$(1, k)$, any $k \geq 1$	3
(j, k), $j \geq 2$	4
$(1, 1, k)$, any $k \geq 1$	5
$(1, j, k)$, $j \geq 2$	6
$(j, 1, k)$, $j \geq 2$	6
$(j, k, 1)$, $j, k \geq 2$	8
(j, k, l), $j, k \geq 2$, $l \geq 2$	8
$(1, 1, 1, k)$	8
$(1, 1, j, k)$, $j \geq 2$	10
$(1, j, 1, k)$, $j \geq 2$	9
$(j, 1, 1, k)$, $j \geq 2$	10
$(1, j, k, l)$, $j, k \geq 2$	12
$(j, 1, k, l)$, $j \geq 2$, $k \geq 2$	12
$(j, k, 1, l)$, $j, k \geq 2$	12
(j, k, l, m), $j, k, l \geq 2$	16

The stabilized values (all parts ≥ 2) by length ℓ are: $1, 2, 4, 8, 16, \ldots = 2^\ell$. This suggests a product structure in the stable regime.

4.2 Profile $c = (2, 0, 0)$, $d = 2$ (second Rogers–Ramanujan)

The stabilized coefficients again follow powers of 2: length ℓ gives $a_\alpha = 2^\ell$ when all parts of α are sufficiently large.

4.3 Profile $c = (2, 1, 1)$, $d = 4$

The coefficients are richer: $a_{(k)} = 5$ for k large; $a_{(j,k)} = 25$ for j, k large; $a_{(j,k,l)} = 125$ for j, k, l large. The stabilized values appear to be 5^ℓ, where 5 is the number of A_2 cylindric partitions of the empty profile (i.e., ordinary plane partitions of size $\leq$ something related to d).

5 Interpretation and structure

5.1 Why quasi-symmetric?

The quasi-symmetry has a natural explanation. Consider a CPP π with frequency composition $\alpha = (a_1, \ldots, a_k)$ at values $v_1 < v_2 < \cdots < v_k$. The

interlacing conditions (1) compare entries at the same positions: $\nu_j^{(i)} \geq \nu_j^{(i')}$ (with shifts). Whether a particular entry equals v_s or $v_{s'}$ matters for the *weight* but the combinatorial constraints depend only on the *relative order* of entries, not on their absolute values.

More precisely: the set of CPPs with value set $\{v_1, \ldots, v_k\}$ and frequency vector $(a_1, \ldots, a_k)$ is in weight-preserving bijection with those using value set $\{w_1, \ldots, w_k\}$ (any other increasing sequence) with the same frequency vector, via the substitution $v_s \mapsto w_s$ applied to every entry. This bijection preserves the interlacing conditions because they only depend on the ordering of entries.

This is precisely the defining property of quasi-symmetry: the coefficient of $z_{v_1}^{a_1} \cdots z_{v_k}^{a_k}$ depends only on $\alpha = (a_1, \ldots, a_k)$, not on $v_1 < \cdots < v_k$.

The argument is an instance of Stanley's P-partition theory [3, 1]: whenever a set of fillings is constrained by conditions that only compare the *relative order* of values (weak and strict inequalities), the generating function tracking value frequencies is quasi-symmetric. The cylindric interlacing conditions (1) are precisely of this form.

Note that the individual $\tilde{g}_n = \sum_{\max(\pi)=n} z_1^{\#_1} \cdots$ (fixed max entry $= n$) is *not* quasi-symmetric: the coefficient of z_n^1 (a single entry equal to n) is positive, while z_1^1 (a single entry equal to 1, but $\max = n$) is zero for $n > 1$. Any constraint on the maximum entry, whether $\max = n$ or $\max \leq m$, breaks quasi-symmetry by privileging certain values over others. Only the unconstrained sum (2) is quasi-symmetric.

Remark 5. *The function is not symmetric because swapping the* multiplicities *(e.g., having a copies of value 1 and b copies of value 2, versus b copies of value 1 and a copies of value 2) changes the interlacing constraints: value 1 appears in inner positions of the interlacing, while value 2 must dominate value 1 entries. The relative* positions *of the values in the interlacing chain distinguish compositions from partitions.*

5.2 Disconnect from the positivity conjecture

The frequency generating function lives in QSym, which is a Sym-module (in fact a Sym-algebra). Natural bases of QSym include the monomial basis $\{M_\alpha\}$ (in which our expansion (3) has manifestly nonneg coefficients), the fundamental basis $\{L_\alpha\}$ of Gessel (see Section 6), and the dual basis to noncommutative symmetric functions.

However, this QSym structure does *not* help with Warnaar's Conjecture 2.7. The conjecture concerns the polynomials $Q_{n,c}(q)$, which are ex-

tracted from the coefficient of z^n in $(zq;q)_\infty \cdot \mathrm{GK}_c(z,q)$. This extraction involves:

1. The max-entry statistic $n(\pi) = \max\{\nu_j^{(i)}\}$, which is *not* a monomial specialization of the z_k (cf. Remark after (2));

2. An alternating-sign sum (the $(zq;q)_\infty$ factor), whose cancellation crosses between different frequency vectors (see Section 7).

The quasi-symmetry of $\widetilde{\mathrm{GK}}_c$ is a property of the *aggregate* over all max entries, but $Q_{n,c}$ lives at a *fixed* max entry. The QSym framework cannot see this decomposition.

In particular, the specialization $z_k \mapsto q^k$ sends $\widetilde{\mathrm{GK}}_c$ to $\mathrm{GK}_c(1,q) = \sum_n g_n(q)$, which is manifestly nonneg since each $g_n(q)$ counts objects. This tells us nothing about the positivity of $Q_{n,c}(q)$, which involves nontrivial cancellation among the g_n.

5.3 Partial quasi-symmetry of $\tilde{g}_n$

Although $\tilde{g}_n$ (CPPs with max $= n$) is not quasi-symmetric in the variables $z_1, z_2, \ldots, z_n$ (since the value n is distinguished), it retains a *partial* quasi-symmetry: the value n is special, but the remaining values $\{1, \ldots, n-1\}$ can be freely relabelled.

More precisely, write

$$\tilde{g}_n = \sum_{m \geq 1} z_n^m \cdot f_{n,m}(z_1, \ldots, z_{n-1}),$$

where $f_{n,m}$ collects those CPPs with exactly m parts equal to n.

Observation 6 (Computational)**.** *For every rank-3 profile c with $d = |c| \leq 5$ and $n = 1, 2, 3$: each slice $f_{n,m}$ is a quasi-symmetric polynomial in $z_1, \ldots, z_{n-1}$. That is,*

$$\tilde{g}_n \ \in \ z_n \cdot \mathrm{QSym}(z_1, \ldots, z_{n-1})[z_n].$$

The argument is the same relabelling bijection as for the full $\widetilde{\mathrm{GK}}_c$, but restricted to non-$n$ values. Given a CPP π with $\max(\pi) = n$ and non-n value set $\{v_1 < \cdots < v_k\} \subset \{1, \ldots, n-1\}$, replacing each v_i by any other increasing sequence $\{w_1 < \cdots < w_k\} \subset \{1, \ldots, n-1\}$ preserves the interlacing conditions: the non-n entries interact with each other only through their relative order, and every non-n entry v satisfies $v < n$, so the inequalities involving n-entries are automatically preserved.

Remark 7 (Failure of slice L-positivity)**.** *The individual slices $f_{n,m}$ are M-positive (the coefficients count CPPs) but are* not *in general L-positive in* $\mathrm{QSym}(z_1, \ldots, z_{n-1})$*. Already for $c = (1, 0, 0)$, $n = 3$, $m = 1$ (two variables z_1, z_2), the M-coefficients $a_{(k)} = 2$ and $a_{(j,k)} = 1$ yield $L_{(a,b)} = M_{(a,b)} - M_{(a+b)} = 1 - 2 = -1$ for every two-part composition. This holds for all 55 tested profiles.*

Remark 8 (Variable-count mismatch)**.** *The E–H factorization $Q_n = (q;q)_n \cdot g_n$ couples two objects that live in different numbers of variables: the factor $(q;q)_n = \prod_{k=1}^{n}(1-q^k)$ lifts to the elementary symmetric functions $e_1, e_2, \ldots$ in* infinitely many *variables, while $\tilde{g}_n$ is a polynomial in only $z_1, \ldots, z_n$. This mismatch is a structural obstacle to any multivariate lift of $Q_{n,c}$: the cancellation producing Q_n from g_n necessarily involves objects living in incompatible numbers of variables.*

5.4 Stabilization and product structure

For $c = (1, 1, 0)$, the stabilized QSym coefficients (all parts of α sufficiently large) satisfy $a_\alpha^{\mathrm{stab}} = 2^{\ell(\alpha)}$, suggesting a product formula

$$\widetilde{\mathrm{GK}}_c^{\mathrm{stab}} = \prod_{k \geq 1} (1 + z_k)^{?}$$

in some asymptotic sense. More generally, for $c = (2, 1, 1)$ with $d = 4$, the stable value is $5^{\ell(\alpha)}$, consistent with the number of maximal cylindric configurations per value. Understanding this product structure is a direction for future work.

6 L-positivity

The *fundamental quasi-symmetric functions* L_α (also denoted F_α in the literature) form a second natural basis of QSym, related to the monomial basis by

$$L_\alpha = \sum_{\beta \preceq \alpha} M_\beta,$$

where $\beta \preceq \alpha$ means β is a refinement of α (obtained by splitting parts of α). Equivalently,

$$L_{(\alpha_1, \ldots, \alpha_k)} = \sum_{\substack{i_1 \leq i_2 \leq \cdots \leq i_n \\ i_j < i_{j+1} \text{ if } j \in D(\alpha)}} z_{i_1} z_{i_2} \cdots z_{i_n},$$

where $n = |\alpha|$ and $D(\alpha) = \{\alpha_1, \alpha_1 + \alpha_2, \ldots, \alpha_1 + \cdots + \alpha_{k-1}\}$ is the descent set of α.

The fundamental basis plays the role in QSym that Schur functions play in Sym: L-positivity is the "right" notion of positivity in QSym. Under the natural map QSym $\to$ Sym that sends $L_\alpha \mapsto s_{\mathrm{sort}(\alpha)}$, L-positivity implies Schur-positivity of the symmetrization.

Observation 9 (Computational)**.** *For every rank-3 profile c with $d = |c| \leq 5$ (55 profiles total), $\widetilde{\mathrm{GK}}_c(\mathbf{z})$ is L-positive:*

$$\widetilde{\mathrm{GK}}_c(\mathbf{z}) = \sum_\alpha b_\alpha(c)\, L_\alpha, \qquad b_\alpha(c) \in \mathbb{Z}_{\geq 0}. \tag{4}$$

Since M-positivity is automatic (the M-coefficients count CPPs), L-positivity is a strictly stronger assertion. The M-to-L conversion uses Möbius inversion on the refinement poset:

$$b_\alpha = \sum_{\beta \succeq \alpha} (-1)^{\ell(\beta)-\ell(\alpha)}\, a_\beta,$$

where $\beta \succeq \alpha$ means α is a refinement of β (equivalently, β is a coarsening of α). The nonnegativity of the b_α after this alternating sum is nontrivial and requires cancellation.

6.1 L-coefficients for $c = (1, 1, 0)$, $d = 2$

Composition α	b_α
$()$	1
(k), any $k \geq 1$	2
$(1, k)$, any $k \geq 1$	1
(j, k), $j \geq 2$	2
$(1, 2, 1)$	1
$(1, j, k)$, $j \geq 2$, $k \geq 2$	1
$(j, 2, k)$, $j \geq 2$	2
(j, k, l), $j, k \geq 2$, $l \geq 2$, $k \geq 3$	2

The L-coefficients are smaller than the M-coefficients and exhibit even cleaner stabilization.

6.2 Combinatorial interpretation

L-positivity has a well-known combinatorial meaning: a quasi-symmetric function is L-positive if and only if it can be written as $\sum_w L_{D(w)}$, a sum of

fundamental quasi-symmetric functions indexed by descent sets, over some set of "words" or "permutations."

Equivalently, by the work of Tewari and van Willigenburg [4], L-positivity of a quasi-symmetric function f is equivalent to f being the quasi-symmetric characteristic of a module over the 0-Hecke algebra $H_n(0)$. Thus Observation 6 implies the existence of a 0-Hecke module whose character is $\widetilde{\mathrm{GK}}_c$.

Remark 10 (The 0-Hecke algebra)**.** *The* 0-Hecke algebra $H_n(0)$ *is the quotient of the free algebra on generators* $T_1, \ldots, T_{n-1}$ *by the Coxeter braid relations together with the* idempotent *relations* $T_i^2 = T_i$ *(in contrast to the symmetric group algebra, where* $T_i^2 = 1$*, or the Hecke algebra* $H_n(q)$*, where* $T_i^2 = (q-1)T_i + q$*). Equivalently,* $H_n(0)$ *is obtained by specializing the Hecke algebra* $H_n(q)$ *at* $q = 0$*.*

The irreducible $H_n(0)$*-modules are one-dimensional and are indexed by compositions of* n *(not by partitions, as for* $\mathfrak{S}_n$*). The quasi-symmetric characteristic* $\mathrm{ch}(M) = \sum L_\alpha$*, summed over composition indices of the irreducible constituents, provides the natural bridge between* $H_n(0)$*-modules and* L*-positive quasi-symmetric functions.*

A related phenomenon appears in the work of Shareshian and Wachs [2], who define a quasi-symmetric refinement of the chromatic symmetric function and prove its L-positivity (and even e-positivity) for certain classes of graphs. Their construction is motivated by the Stanley–Stembridge conjecture and is analogous in spirit: a combinatorial generating function, refined by a "descent-like" statistic, turns out to be L-positive.

Conjecture 11. *There exists a natural descent-type statistic on cylindric partitions such that* $\widetilde{\mathrm{GK}}_c = \sum_\pi L_{\alpha(\pi)}$*, where* $\alpha(\pi)$ *is the descent composition of* π*.*

Finding such a statistic would give a combinatorial proof of L-positivity and could shed light on the original positivity conjecture for $Q_{n,c}$.

6.3 Failure of the E–H factorization in QSym

The Euler–Hamburger factorization $\mathrm{GK}_c = H \cdot F_c$, where $H = 1/(zq;q)_\infty$ and $F_c = (zq;q)_\infty \cdot \mathrm{GK}_c$, is the key identity connecting GK_c to the polynomials $Q_{n,c}$.

We tested two natural QSym analogs:

1. **With max-entry variable.** Introduce a variable t tracking the maximum entry and define $\widetilde{\mathrm{GK}}_c(t, \mathbf{z}) = \sum_n t^n \tilde{g}_n$, then compute $[t^n]\big(\prod_{k\geq 1}(1-$

$tz_k) \cdot \widetilde{\mathrm{GK}}_c(t, \mathbf{z})\big) = \sum_{j=0}^{n} (-1)^j e_j \cdot \tilde{g}_{n-j}$. This is *not* quasi-symmetric and has *negative* coefficients, because the individual $\tilde{g}_n$ (CPPs with max entry n) are not quasi-symmetric: fixing the max entry privileges a particular value.

2. **Without max-entry variable.** Compute $\prod_{k\geq 1}(1 - z_k) \cdot \widetilde{\mathrm{GK}}_c(\mathbf{z})$ directly in QSym, which under $z_k \mapsto q^k$ specializes to $(q;q)_\infty \cdot \mathrm{GK}_c(1, q)$. This also has *negative* coefficients and is not quasi-symmetric.

Both failures indicate that the E–H factorization is fundamentally tied to the max-entry variable z, which does not sit naturally in the QSym framework.

7 Attempt to lift Q_n to QSym: negative coefficients

The positive results above concern the *full* generating function $\widetilde{\mathrm{GK}}_c$, which sums over all cylindric partitions without constraint on the maximum entry. The central positivity problem, however, is about the *polynomials* $Q_{n,c}(q)$, which are extracted by the alternating-sign formula

$$Q_{n,c}(q) = \sum_{j=0}^{n} \sum_{\substack{S\subseteq\{1,\ldots,n\} \\ |S|=j}} \sum_{U\subseteq\{1,\ldots,n-j\}} \sum_{\substack{\nu\in\mathrm{CPP}(c) \\ \max(\nu)=n-j}} (-1)^{|S|+|U|}\, q^{\sigma(S)+\sigma(U)+|\nu|}. \tag{5}$$

The natural question is: can the QSym machinery, which works so well for $\widetilde{\mathrm{GK}}_c$, be applied to $Q_{n,c}$?

7.1 The QSym Q-polynomial

Given a triple (S, U, ν) from the ground set $\widetilde{\mathcal{G}}_{n,c}$, write $m = n - |S|$ and define the *combined frequency vector*:

$$\mathrm{freq}(S, U, \nu)_k = |\{s \in S : s = k\}| + |\{u \in U : u = k\}| + |\{(i, j) : \nu_j^{(i)} = k\}|,$$

counting how many times the value k appears across all three components. The total weight is $\sigma(S) + \sigma(U) + |\nu| = \sum_k k \cdot \mathrm{freq}(S, U, \nu)_k$, so replacing $q^k \mapsto z_k$ in each factor gives a quasi-symmetric refinement:

$$Q_n^{\mathrm{QSym}} = \sum_{(S,U,\nu)\in\widetilde{\mathcal{G}}_{n,c}} (-1)^{|S|+|U|}\, M_{\mathrm{freq}(S,U,\nu)}. \tag{6}$$

Under the principal specialization $z_k \mapsto q^k$, $M_\alpha \mapsto \sum_{i_1<\cdots<i_\ell} q^{\sum a_s i_s}$, and summing over all frequency vectors recovers $Q_{n,c}(q)$.

7.2 Failure: negative M-coefficients

Observation 12 (Computational)**.** *For $n \geq 2$, the QSym polynomial Q_n^{QSym} has* negative *coefficients in the monomial basis M_α. This holds for every tested profile with $d \leq 5$, including the simplest case $c = (1, 0, 0)$, $d = 1$.*

Concretely, for $c = (1, 1, 0)$, $n = 2$:

$$Q_2^{\mathrm{QSym}} = M_{(1,1)} - 2\, M_{(2)} + M_{(1,1,1)} - M_{(2,1)} + \cdots$$

The coefficient $a_{(2)} = -2$ is negative. Since M-positivity fails, L-positivity fails *a fortiori*: the L-basis expansion also contains negative terms.

Remark 13. *For $n = 1$, Q_1^{QSym} is M-positive and in fact equal to h_1^{QSym}, where h_m is the "corrected" generating function $h_m = (q; q)_m \cdot g_m$ (cf. Section 6). The failure begins precisely at $n = 2$, where the alternating q-binomial sum introduces cancellation between different frequency vectors.*

7.3 Anatomy of the failure

The key issue is *cross-cancellation*: different frequency vectors can map to the same q-weight under specialization, so cancellation in $Q_n(q)$ *must* cross QSym monomials.

More precisely, for a given q-weight w, define:

- The *internal cancellation* at frequency α: positive and negative triples with freq $= \alpha$ and total weight w that cancel within the same QSym class.
- The *cross-cancellation*: excess positive contributions from one frequency α cancelling excess negative from another frequency β, where both have $|\alpha| = |\beta| = w$ (same total weight, different composition).

Computationally, approximately 94–97% of cancellation is internal (same frequency vector, handled by the U-toggle involution). But the remaining 0.3–3% is cross-cancellation, which is *essential*: the net coefficient at some individual frequency vectors is negative, and only the aggregation across all compositions of the same weight yields the positive Q_n coefficient.

Example 14. *For $c = (1, 1, 0)$, $n = 2$, at q-weight 4: the frequency vector $(0, 2)$ (two copies of value 2) has net coefficient -1, while $(1, 0, 1)$ (one copy of value 1 and one of value 3) has net $+2$. Under $z_k \mapsto q^k$, both map to q^4, and their contributions $-1 + 2 = 1$ yield $[q^4]\, Q_2 = 1 \geq 0$.*

7.4 The "ν-only" variant

One might hope that tracking only the CPP parts ν in the frequency vector (keeping S and U as scalar q-factors) would avoid the problem:

$$Q_n^{\nu\text{-only}} = \sum_{(S,U,\nu)} (-1)^{|S|+|U|}\, q^{\sigma(S)+\sigma(U)} \cdot M_{\mathrm{freq}(\nu)}. \tag{7}$$

This is a hybrid object: a formal power series in q with coefficients in QSym.

Observation 15 (Computational)**.** *The ν-only variant* (7) *also has negative coefficients, and in fact fails* worse *than the combined version* (6)*: it is negative already at $n = 1$, for every profile tested.*

The failure at $n = 1$ is structural. When $n = 1$, the ground set includes $(S = \{1\}, U = \emptyset, \nu = \emptyset)$ with sign -1 and q-scalar q^1, contributing $-q \cdot M_{()}$ to $Q_1^{\nu\text{-only}}$. Meanwhile the positive contributions from $(S = \emptyset, U, \nu)$ are spread across many frequency vectors M_α with $\alpha \neq ()$. There is no way for these to cancel the $-q \cdot M_{()}$ term, since it lives in a different QSym component.

By contrast, the combined version (6) works at $n = 1$ precisely because $S = \{1\}$ contributes to $M_{(1)}$ (the same QSym monomial as a CPP part equal to 1), enabling internal cancellation.

7.5 Diagnosis

The QSym framework succeeds brilliantly for the *unconstrained* generating function $\widetilde{\mathrm{GK}}_c$, where every coefficient is manifestly nonneg (it counts CPPs). The difficulty arises when we try to *extract* Q_n, which requires:

1. The max-entry constraint $\max(\nu) = m$ (privileges a particular value, breaking the quasi-symmetric relabelling argument);

2. The alternating sign $(-1)^{|S|+|U|}$ (introduces cancellation that crosses between different frequency vectors).

The q-specialization $z_k \mapsto q^k$ "forgets" the composition structure and allows cross-frequency cancellation to produce the positive polynomial $Q_n(q)$. This forgetting is essential and cannot be avoided within QSym.

In summary: $\widetilde{\mathrm{GK}}_c \in \mathrm{QSym}$ with L-positivity is a genuine structural property of cylindric partitions, but it does not extend to the Q-polynomials. The positivity of $Q_{n,c}(q)$ appears to be an *intrinsically one-variable* phenomenon.

8 Computational verification

The quasi-symmetry was verified as follows. For each profile c with $d \leq 5$:

1. Enumerate all CPPs with maximum entry $\leq n_{\max}$ and total weight $\leq w_{\max}$, using the shifted dominance conditions (1).
2. For each CPP, compute the frequency vector $(\#_1, \#_2, \ldots, \#_{n_{\max}})$ and extract the composition α (ordered nonzero entries).
3. Group CPP counts by composition and verify that all placements of the same composition at different value positions give the same count (restricting to compositions whose weight does not exceed $w_{\max}$ at any placement, to avoid truncation artifacts).

$d = \lvert c \rvert$	Profiles tested	QSym failures	L-negative terms
1	3	0	0
2	6	0	0
3	10	0	0
4	15	0	0
5	21	0	0
Total	**55**	**0**	**0**

Parameters used: $n_{\max} = 3$, $w_{\max} = 12$, with 24 non-truncated quasi-symmetry checks per profile. The L-basis conversion uses Möbius inversion on the refinement poset of compositions.

The partial quasi-symmetry of $\tilde{g}_n$ (Observation **??**) was verified separately by fixing the max entry n and checking that each z_n^m slice is quasi-symmetric in $z_1, \ldots, z_{n-1}$:

$d = \lvert c \rvert$	Profiles tested	Partial QSym failures	Slice L-positive
1	3	0	No
2	6	0	No
3	10	0	No
4	15	0	No
5	21	0	No
Total	**55**	**0**	—

Parameters: $n = 1, 2, 3$, $w_{\max} = 18$. Partial quasi-symmetry holds for all 165 test cases (55 profiles $\times$ 3 values of n). Slice L-positivity fails at $n = 3$ for every profile (the slices $f_{n,m}$ have M-coefficients that decrease with composition length, forcing negative L-coefficients).

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